

Esercizi

1. Calcolare l'area del segnale a tempo continuo $x(t) = \text{sinc}^4(2t - 1)$.

2. Calcolare la trasformata di Fourier del segnale a tempo discreto

$$x(nT) = \begin{cases} \cos(2\pi f_0 nT), & n = 0, \dots, N-1, \\ 0, & \text{altrimenti.} \end{cases}$$

3. Calcolare la convoluzione $z(t) = x_1 * x_2 * x_3(t)$ dei segnali a tempo continuo $x_1(t) = V_1 \text{sinc}(2t/T)$, $x_2(t) = V_2 \text{sinc}^2(t/T)$, $x_3(t) = -V_3 \delta_R(t + 4t_0)$.

4. Calcolare la trasformata di Fourier del segnale a tempo continuo

$$x(t) = \sum_{k=-\infty}^{+\infty} \text{rect}\left(\frac{t-2kT}{T}\right) - \sum_{k=-\infty}^{+\infty} \text{rect}\left(\frac{t-T+2kT}{T}\right).$$

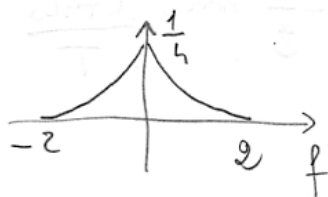
5. Calcolare la risposta impulsiva del filtro, con ingresso $a(kT)$ e uscita $b(kT)$, realizzato mediante l'equazione alle differenze

$$b(kT) = a(kT) + 0.1 b(kT - T) - 0.01 b(kT - 2T).$$

1. Area ($\text{sinc}^4(2t-1)$) = Area ($\text{sinc}^4 2t$)

(invarianza rispetto alla traslazione)

$$\int_{-\infty}^{+\infty} \text{sinc}^2(2t) \text{sinc}^2(2t) dt = \int_{-\infty}^{+\infty} \frac{1}{2} \left(\text{triangolo } \frac{f}{2} \right) \cdot \frac{1}{2} \left(\text{triangolo } \frac{f}{2} \right) df$$



Parseval

$$= 2 \cdot \frac{1}{4} \cdot 2 \cdot \frac{1}{3} = \frac{1}{3}$$

2. $x(nT) = s(nT) p(nT)$

$$s(nT) = \cos(2\pi f_0 nT)$$

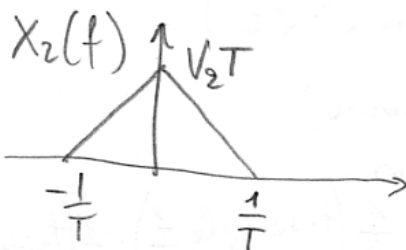
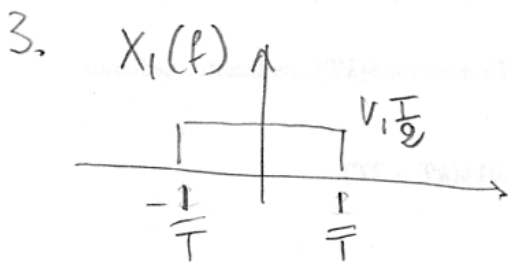
$$p(nT) = \begin{cases} 1 & n = 0 \dots N-1 \\ 0 & \text{altrimenti} \end{cases}$$

$$X(f) = \{S(f)\} * \{P(f)\}$$

$$S(f) = \frac{1}{2} \delta_{\frac{1}{2T}}(f-f_0) + \frac{1}{2} \delta_{\frac{1}{2T}}(f+f_0)$$

$$P(f) = NT \operatorname{sinc}_N(fNT) e^{-j2\pi \frac{N-1}{2} fT}$$

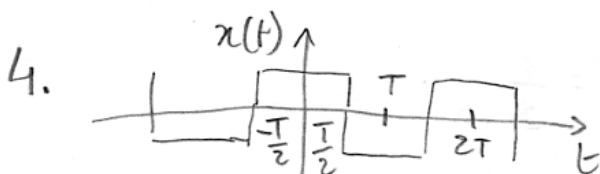
$$X(f) = \frac{1}{2} NT \left(\operatorname{sinc}_N((f-f_0)NT) e^{-j2\pi \frac{N-1}{2} (f-f_0)T} + \operatorname{sinc}_N((f+f_0)NT) e^{-j2\pi \frac{N-1}{2} (f+f_0)T} \right)$$



$$X_1(f) \cdot X_2(f) = \frac{V_1 T}{2} X_2(f)$$

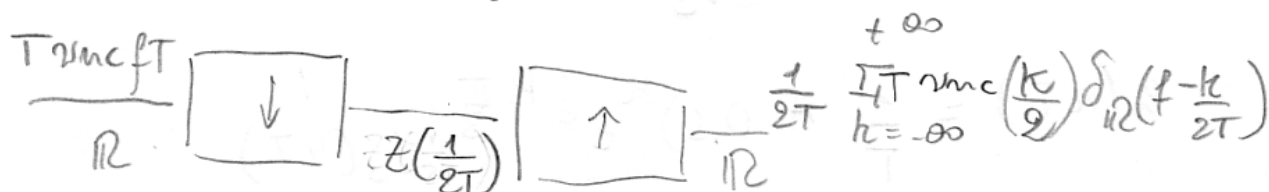
$$x_1 * x_2(t) = \frac{V_1 T}{2} V_2 \operatorname{sinc}^2 \frac{t}{T}$$

$$x_1 * x_2 * x_3(t) = -\frac{V_1 V_2 V_3 T}{2} \operatorname{sinc}^2 \frac{t+t_0}{T}$$



$$x(t) = \sum_{n=-\infty}^{+\infty} \operatorname{rect}\left(\frac{t-2nT}{T}\right)$$

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$$X(f) = \sum_{k=-\infty}^{+\infty} x_{mc} \frac{k}{2} \delta_R\left(f - \frac{k}{2T}\right) - \delta_R(f)$$

5.

$$\frac{B(f)}{A(f)} = \frac{1}{1 - 0.1e^{-j2\pi fT} + 0.01e^{-j4\pi fT}}$$

Poli: radici di

$$z^2 - 0.1z + 0.01$$

$$P_{1,2} = \frac{0.1 \pm \sqrt{0.01 - 4 \cdot 0.01}}{2}$$

$$= 0.1 \left(\frac{1}{2} \pm j \frac{\sqrt{3}}{2} \right) = 0.1 e^{\pm j \frac{\pi}{3}}$$

$$H_z(z) = \frac{1}{(1 - 0.1e^{j\frac{\pi}{3}}z^{-1})(1 - 0.1e^{-j\frac{\pi}{3}}z^{-1})}$$

$$= \frac{A}{(1 - 0.1 e^{j\frac{\pi}{3}} z^{-1})} + \frac{B}{1 - 0.1 e^{-j\frac{\pi}{3}} z^{-1}}$$

$$A = \frac{1}{2} - j \frac{1}{2\sqrt{3}}$$

$$B = A^* = \frac{1}{2} + j \frac{1}{2\sqrt{3}}$$

$$h(kT) = \frac{1}{T} A (0.1)^k e^{j k \frac{\pi}{3}} 1_0(kT)$$

$$+ \frac{1}{T} A^* (0.1)^k e^{-j k \frac{\pi}{3}} 1_0(kT)$$

$$= \frac{2}{T} (0.1)^k \operatorname{Re} \left(A e^{j k \frac{\pi}{3}} \right) 1_0(kT)$$

$$= \frac{2}{T} (0.1)^k \left(\frac{1}{2} \cos k \frac{\pi}{3} + \frac{1}{2\sqrt{3}} \sin k \frac{\pi}{3} \right) 1_0(kT)$$